6

223611496

8

223511323

5

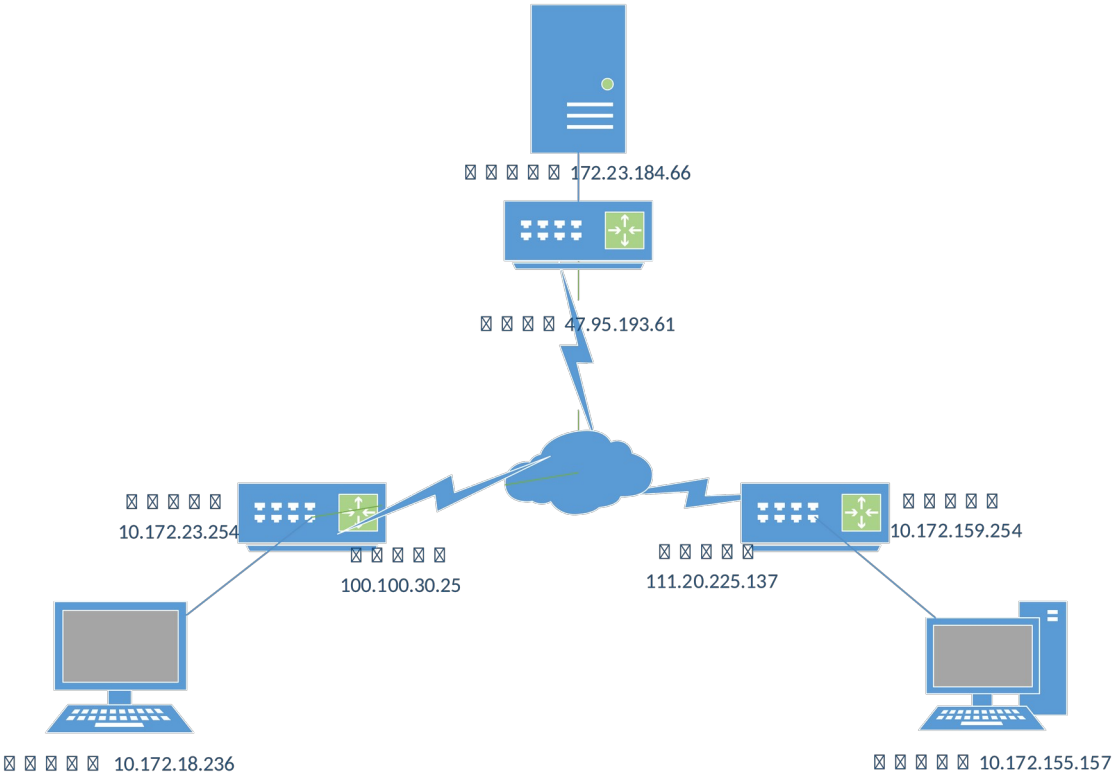
6 TCP

6.1

1. TCP TCP TCP TCP
2. TCP TCP
3. TCP TCP
4. TCP TCP
5. TCP

6.2

1. TCP TCP
2. TCP TCP TCP TCP MSS SACK Window Scale Timestamp
3. RTT MSS
4. TCP
5. TCP



6.3

6.3.1

2

6.3.2 TCP

1 PC2 web Apache IIS Python WEB 5MB

	172.23.184.66
	47.95.193.61
	pic.zip
	3.98 MB

2 PC1 Linux tcpdump

3 PC1 实验 **2** 编写的 **HTTP** WEB

3.1 实验 **2** 编写的 **HTTP** ms WEB 5MB

```
HTTP

    : 47.95.193.61
  : /pic.zip
  : 80
       IPv4    : 47.95.193.61
DNS     : 5.659  ms
TCP          : 25.624  ms
        : 1439.249  ms
HTTP      : HTTP/1.0  200  OK
HTTP      : 200
        : 4170186
         : 4169989
        : pic.zip

=====       HTTP      =====
GET /pic.zip HTTP/1.1
Host: 47.95.193.61
User-Agent: Huang YanShan
Accept: text/html,*/*
Accept-Encoding: identity
Connection: close


=====       =====
HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.10.12
Date: Fri, 17 Apr 2026 02:24:51 GMT
Content-type: application/zip
Content-Length: 4169989
Last-Modified: Fri, 17 Apr 2026 02:12:50 GMT
```

3.2 HTTP □□□□

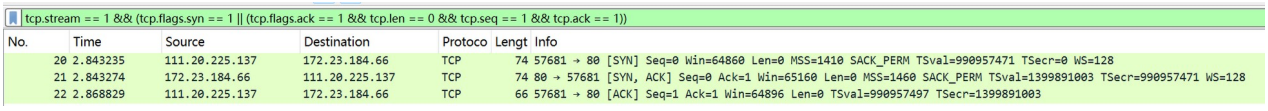
```
HTTP

    : 47.95.193.61
  : /pic.pcap
  : 80
       IPv4    : 47.95.193.61
DNS     : 2.064  ms
TCP          : 27.595  ms
        : 2186.886  ms
HTTP      : HTTP/1.0  200  OK
HTTP      : 200
        : 4475299
         : 4475089
        : pic.pcap

=====       HTTP      =====
GET /pic.pcap HTTP/1.1
Host: 47.95.193.61
User-Agent: Huang YanShan
Accept: text/html,*/*
Accept-Encoding: identity
Connection: close


=====       =====
HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.10.12
Date: Fri, 17 Apr 2026 02:26:58 GMT
Content-type: application/vnd.tcpdump.pcap
Content-Length: 4475089
Last-Modified: Fri, 17 Apr 2026 02:24:56 GMT
```

4 TCP MSS SACK Window Scale Timestamp



4.1 MSS SACK Window Scale TCP

TCP	
MSS	TCP IP
SACK	
Window Scale	TCP header 16

4.2 PC1 TCP MSS SACK Window Scale

TCP		
MSS	1410	1460
SACK	Permitted	Permitted
Window Scale	7	7
	128	128

5 TCP TCP RTT

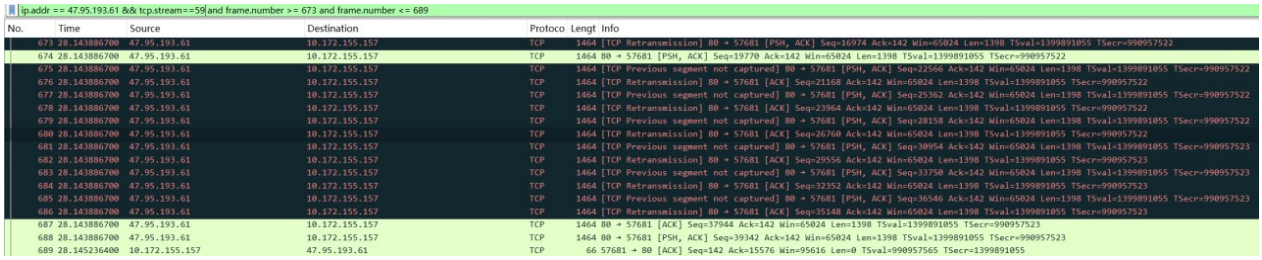
5.1 相关 TCP 过滤器＋分组列表，列表中只有三次握手的这 **3** 个 **TCP** 分组）

5.2 TCP RTT ms

RTT = t(22) - t(21)
= 2.868829 s - 2.843274 s
= 0.025555 s
= 25.555 ms

6 5MB **TCP** 流 务必从建立连接开始抓包 TCP

6.1 相关 TCP 10-15 错误恢复 1 过滤器＋分组列表，列表中只给出填表中需要的 **TCP** 分组）



NO.672 NO.673 NO.691

ACK 3156457429 “ ” NO.673

NO.692 3156460225 TCP

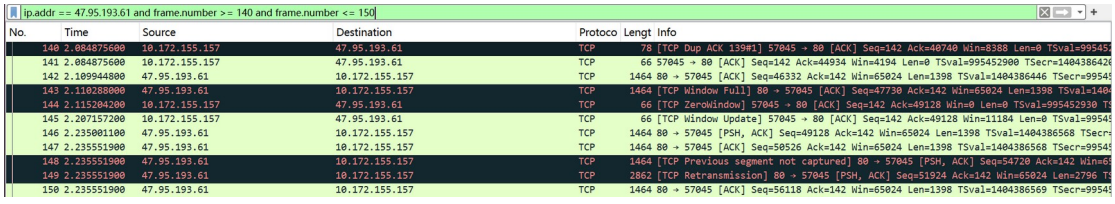
NO.672 NO.693 NO.674 RTO

6.2 □□□□ □□□□ □□□□ 10-15

No.	Seq Number	Ack Number			( )
670	3156454633	1678115225	508	1398	689
671	3156456031	1678115225	508	1398	690
672	3156458827	1678115225	508	1398	692
673	3156457429	1678115225	508	1398	692
674	3156460225	1678115225	508	1398	693
689	1678115225	3156456031	747	0	670
690	1678115225	3156457429	769	0	671
691	1678115225	3156457429	769	0	671
692	1678115225	3156460225	753	0	672 673
693	1678115225	3156461623	743	0	674

6.3 □□ TCP 9-13 □□□□ 1 □□

□+□□□□□□□□□□□□□□□□□ **TCP** □□□



6.4 □□□□

No.	Seq Number	Ack Number			( )
140	142	40740	8388	0	133
141	142	44934	4194	0	134

142	46332	142	508	1398	89
143	47730	142	508	1398	89
144	142	49128	0	0	143
145	142	49128	11184	0	143
146	49128	142	508	1398	89
147	50526	142	508	1398	89
148	54720	142	508	1398	89

6.5 6.3

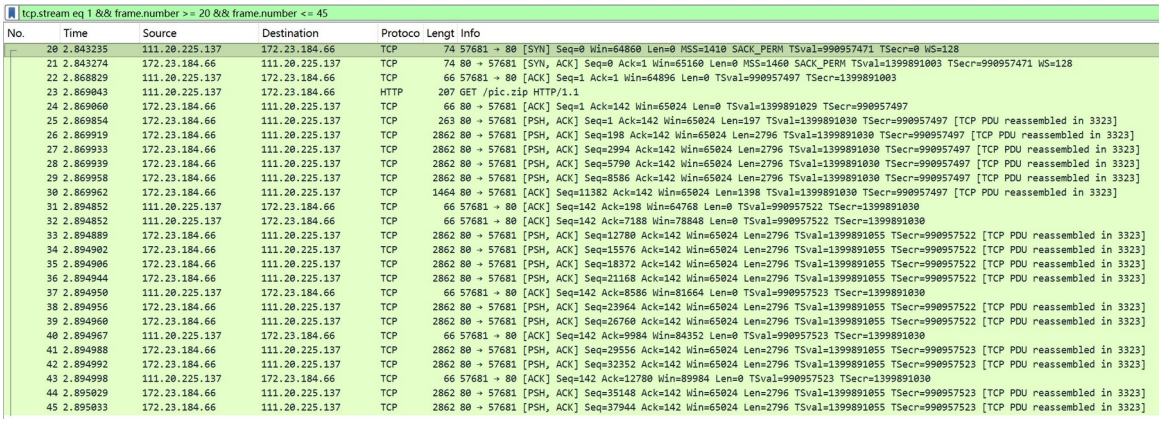
			ACK			140	8388
141	4194				144	0	Zero
Window				145	11184	146	147
148							

7 TCP

cwnd

7.1 TCP □□□ 1 □□□□□□□□□□□ □ **TCP** □

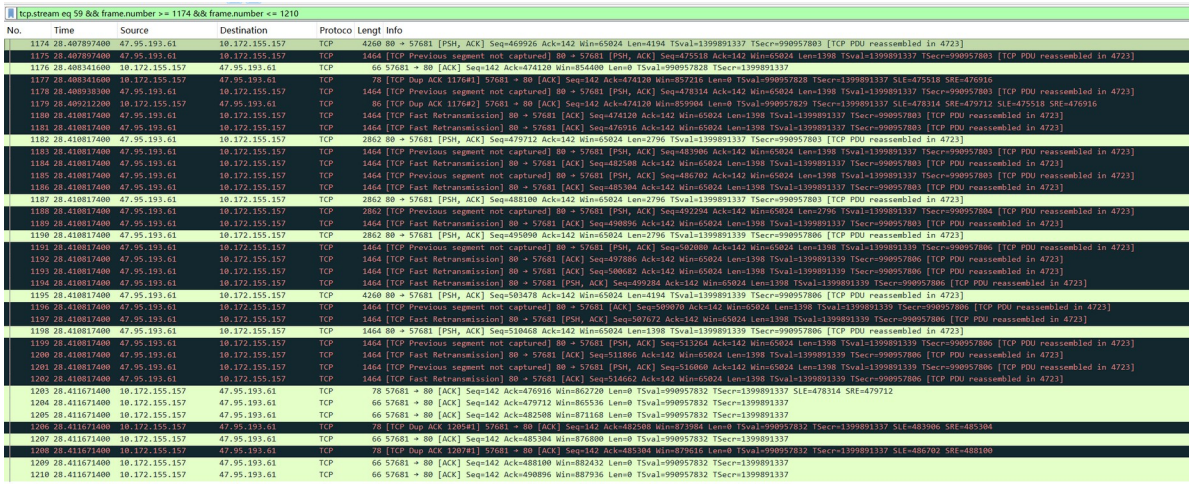
□□



7.2 □□□

1. 22 t=2.868829s HTTP 23
2. 25-30 ACK 6
   25: seq=1, len=197 HTTP
   26: seq=198, len=2796 = 2×1398 TSO 2 MSS
   27: seq=2994, len=2796
   28: seq=5790, len=2796
   29: seq=8586, len=2796
   30: seq=11382, len=1398
   197+2796×4+1398 = 12779 = 10 MSS cwnd
3. ACK 31-32 t=2.894852s 25ms ≈1 RTT
   ACK cwnd 1 MSS
4. 33-45 cwnd 20 MSS 27960

7.3 TCP □□□□ 1 □□□□□□□□□□□ □ **TCP** □□□



7.4 TCP □□□□

1.

1174 seq=469926, len=4194 469926-474119

1175 seq=475518, len=1398 475518-476915

474120-475517 1 MSS

2. ACK

1176 ACK ack=474120 474120

1177 ACK ack=474120 [TCP Dup ACK #1] 474120

1179 ACK ack=474120 [TCP Dup ACK #2] 474120

3.

1180 seq=474120, len=1398 [TCP Fast Retransmission]

474120-475517

1181-1202

4.

1203 ACK ack=476916, 479712 ...

7.5 cwnd MSS

TCP cwnd MSS	10

6.3.3 TCP

8 TCP

TCP

9 8

10 500KB

11 TCP

6.4

12

12.1 TCP

TCP 发送方 ACK cwnd ssthresh

12.2 TCP

TCP 接收方 ACK rwnd

12.3 TCP

TCP 面向连接、可靠、按字节流传输 TCP TCP

TCP TCP